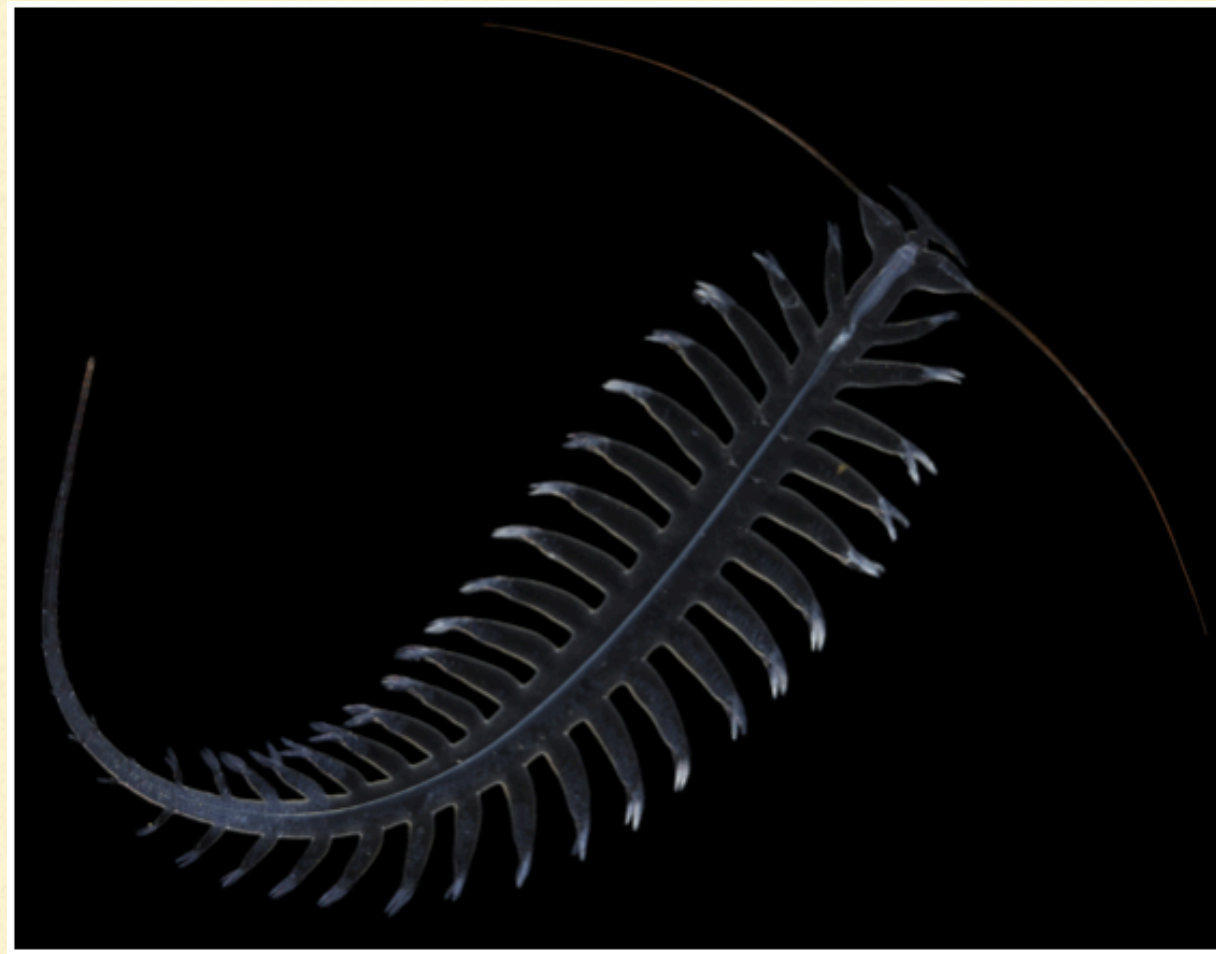


Study of photoperception and regulation of bioluminescence in the marine polychaete *Tomopteris helgolandica* (Tomopteridae)



Supervisor: Professor Jérôme Mallefet
Advisor: Dr Laurent Duchatelet

Bioluminescence:

Production of visible light by a living organism through a biochemical reaction.

(Harvey, 1952)

Different functions:

(Morin, 1983, Haddock et al., 2010)

Predation



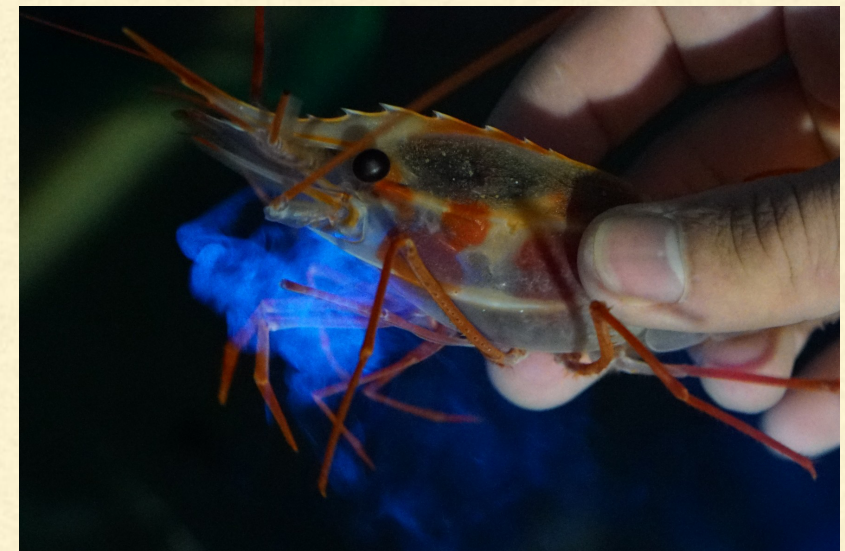
Genus Lophiiformes

Communication



Ostracods

Defense



Shrimp
Heterocarpus sibogae

Nervous :

Acetylcholine



Annelids

Echinoderms

(Mallefet et al., 2004 ; Anctil et al., 1989)

- homo- or hetero-pentamer

α

α et β

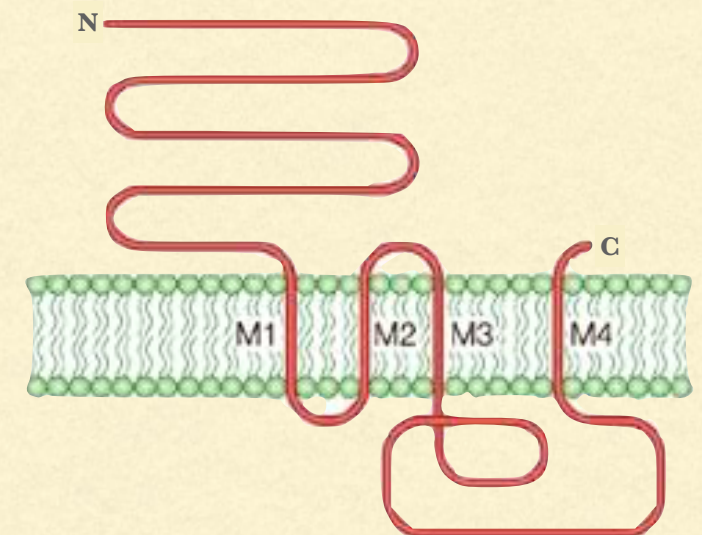
Most conserved subunit across the animal kingdom

- subunit $\alpha 7$

Controls:

Nicotinic receptors :

- 5 subunits



Nervous :

Acetylcholine



Annelids

Echinoderms

(Mallefet et al., 2004 ; Anctil et al., 1989)

Hormonal :



Chordata: *E. molleri* et *E. spinax*

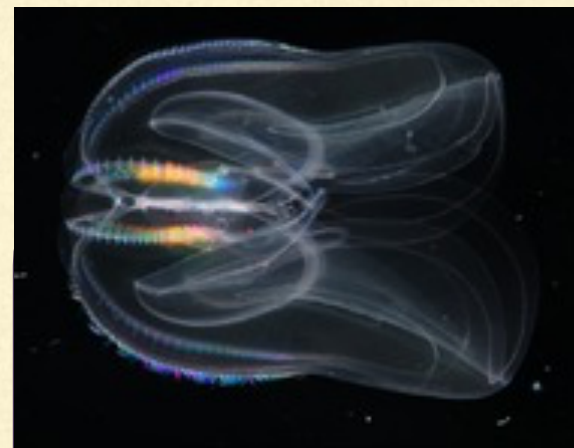
(Duchatelet et al., 2020 ; Claes & Mallefet, 2015)

Controls:

Feed back :

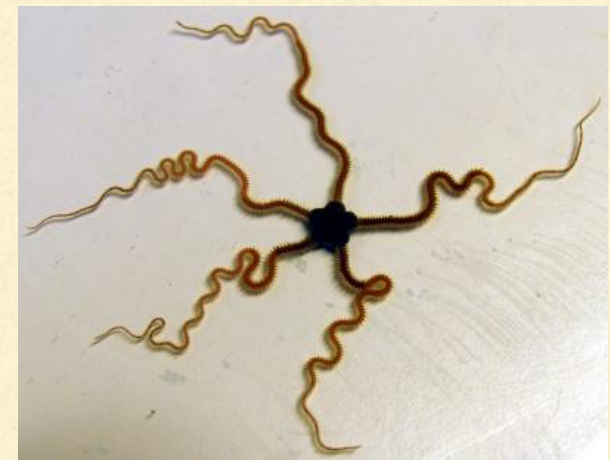
Euprymna scolopes

(Tong et al., 2009)



Mnemiopsis leidyi

(Schnitzler et al., 2012)



Amphiura filiformis

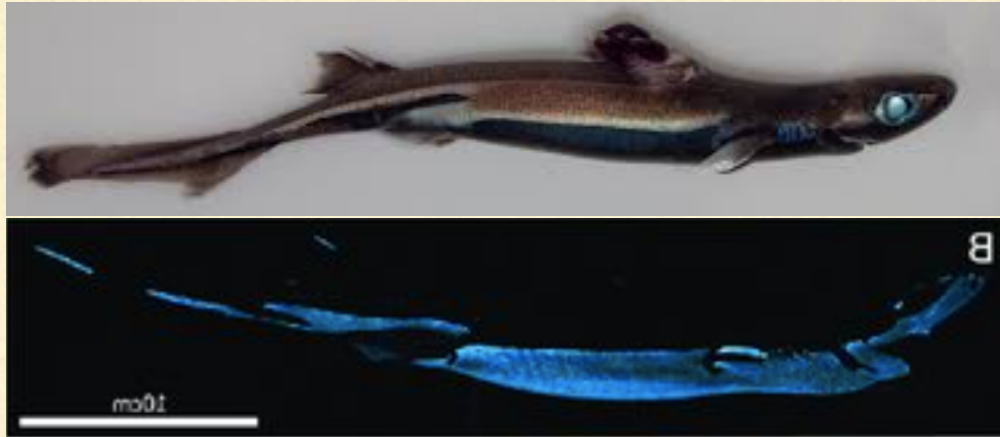
(Delroisse et al., 2014)



Janicella spinicauda

(Bracken-Grissom et al., 2020)

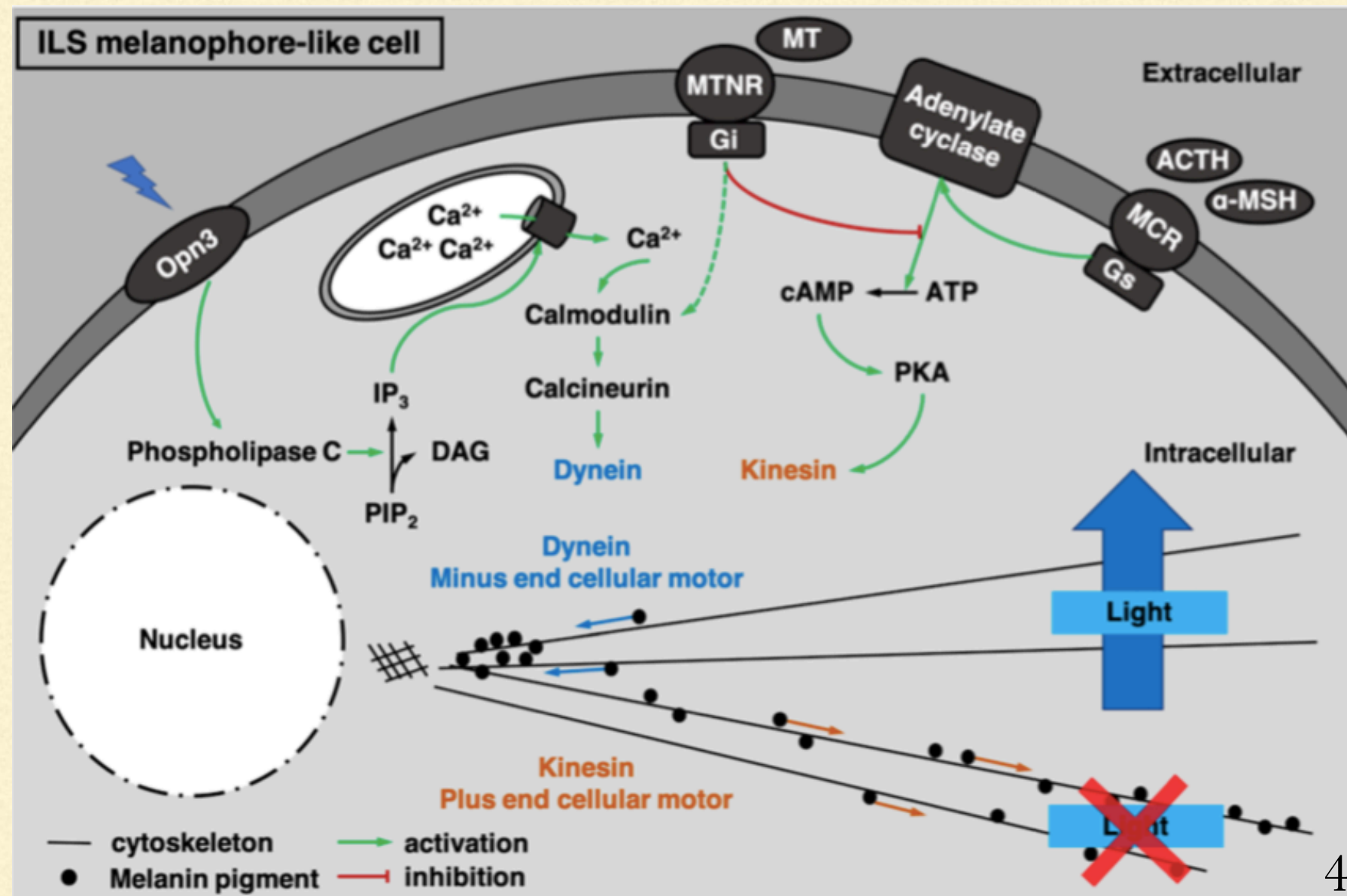
Feed back :



Etmopterus spinax

(Duchatelet et al., 2020)

Demonstration of the involvement of an opsin in the feedback regulation of bioluminescence.



Photosensitive protein :

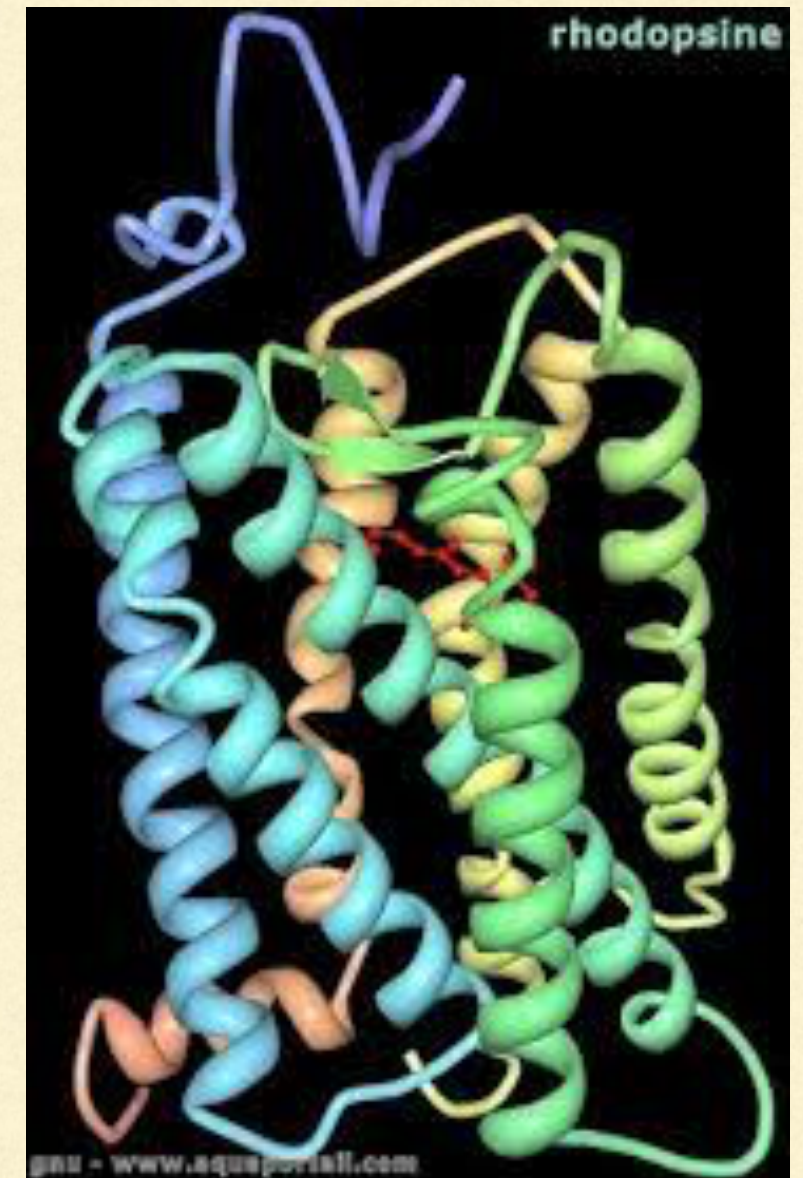
Photoreceptor protein + chromophore
opsin retinal

Retinal :

Conformational change
11-cis $\longrightarrow$ all-trans

Melanopsin :

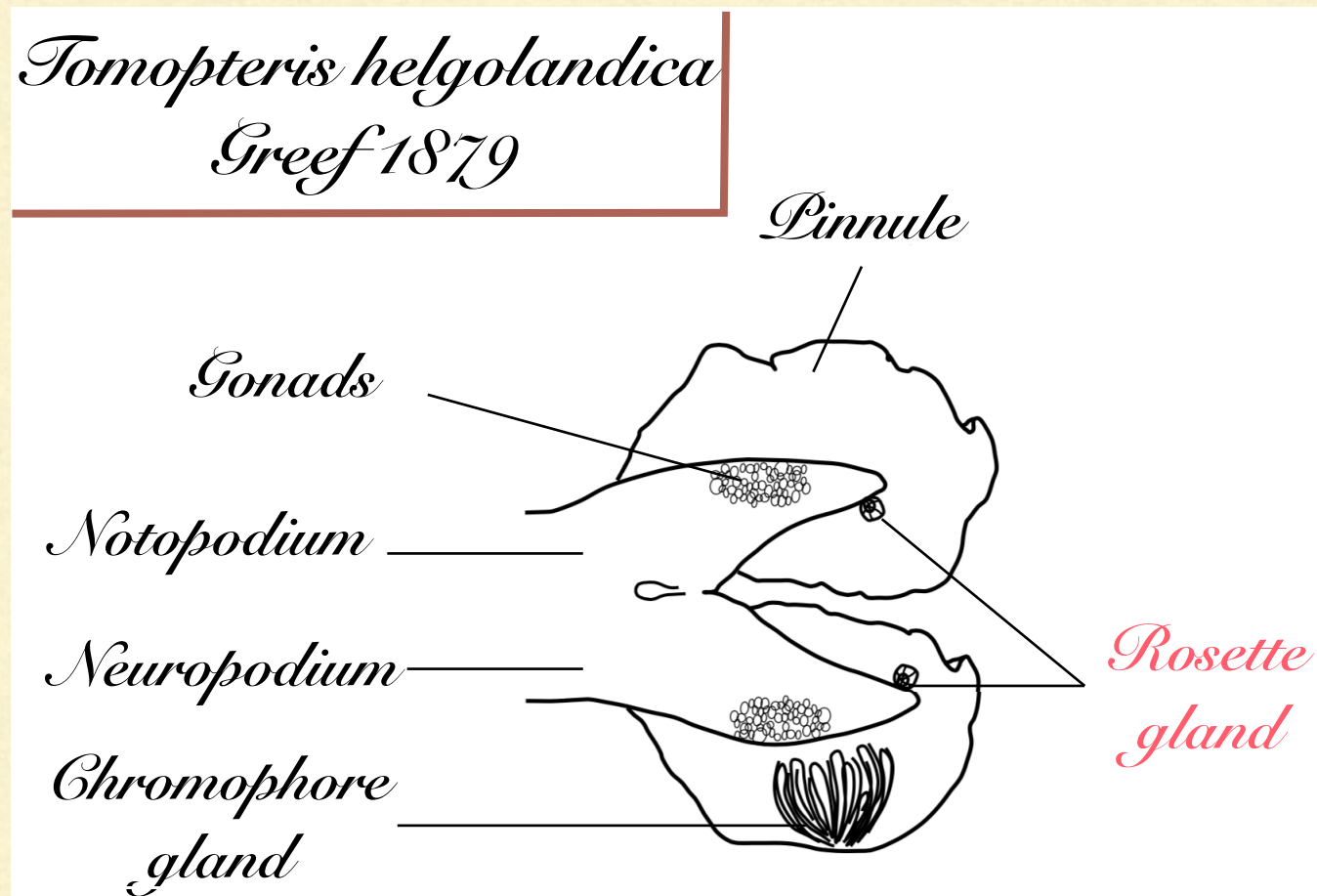
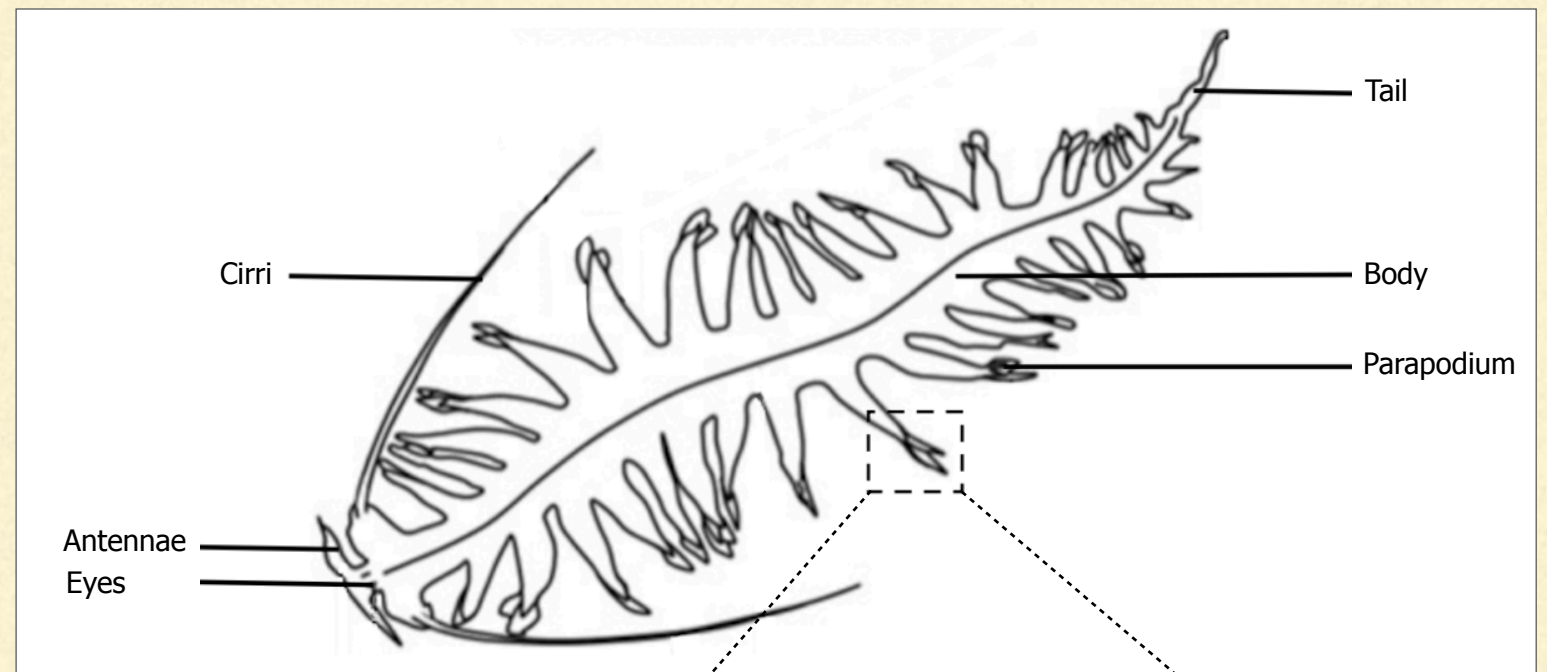
Opsin 4 $\longrightarrow$ Gq protein-coupled receptor



Tomopteris helgolandica :



(Gouveneaux & Mallefet, 2013)



(Gouveneaux, 2016)



Main Objective :

The main objective of this thesis is to further our understanding of photoperception and the regulation of light emission in *Tomopteris helgolandica*.

- ✱ (i) Does *T. helgolandica* possess the molecular components involved:
 - (ia) in the control of light emission via acetylcholine ?
 - (ib) in photoperception via opsin 4 ?

- ✱ (ii) Where are the main protein components involved in the two photobiological processes located in *T. helgolandica*, namely the nicotinic receptors and opsin 4?

Materials and Methods :

In silico transcriptomic analyses:

- > Opsin 4 signaling cascade
- > Nicotinic receptor signaling cascade
- > Antibodies targeting the $\alpha 7$ subunit of the acetylcholine receptor

Western blot:

- > Presence of nicotinic receptors throughout the body

Immunolabeling:

- > Localization of opsin 4
- > α -tubulin staining
- > Localization of nicotinic receptors

Results and discussion :

In silico: transcriptomic analysis

Presence of nearly all proteins involved in the signaling cascades:

- Opine 5 (15/16)
 - Nicotinic receptors (14/18)
-

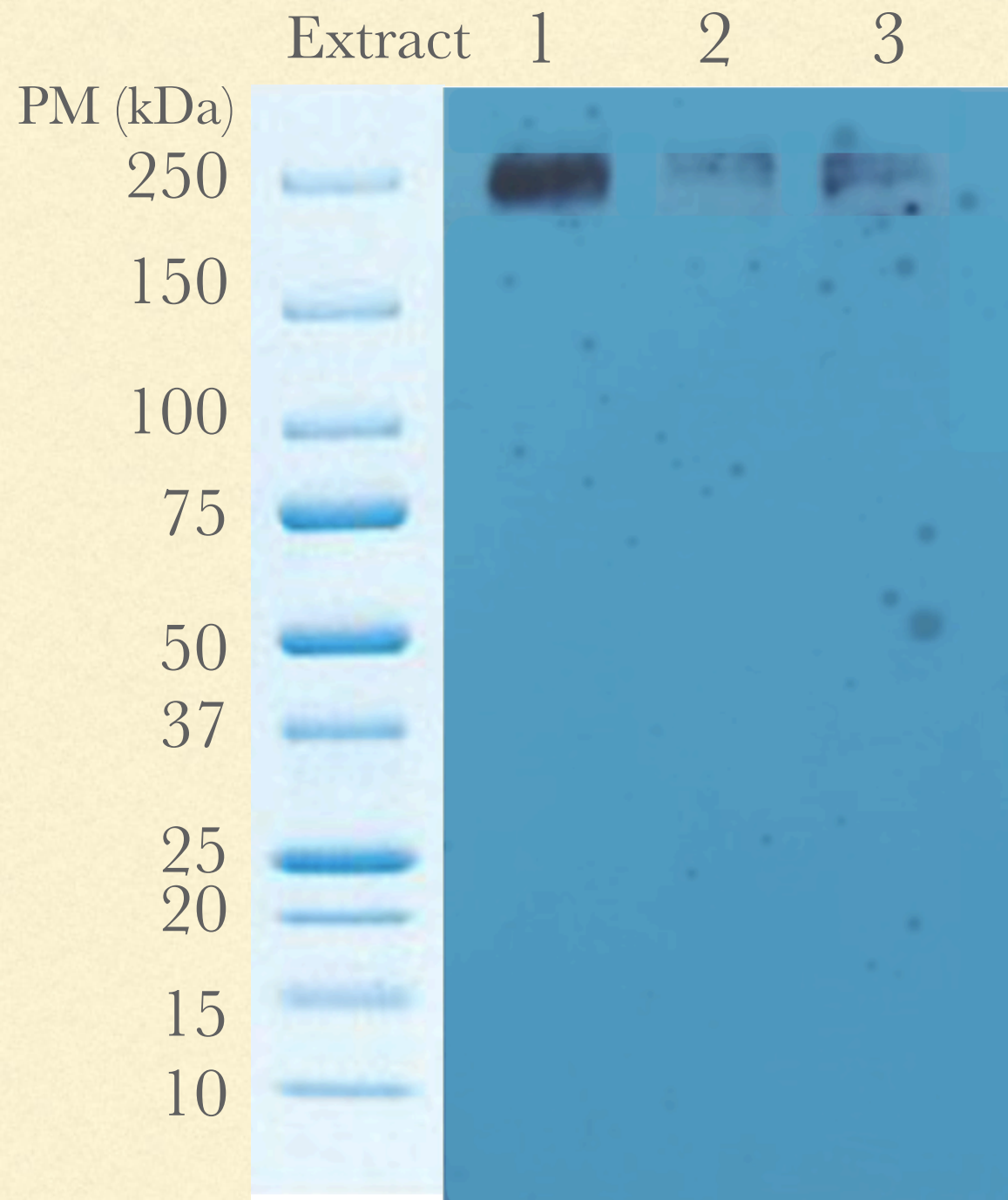


Comparison of the $\alpha 7$ subunit sequence of human neuronal nAChR with that of *T. helgolandica* nAChR.

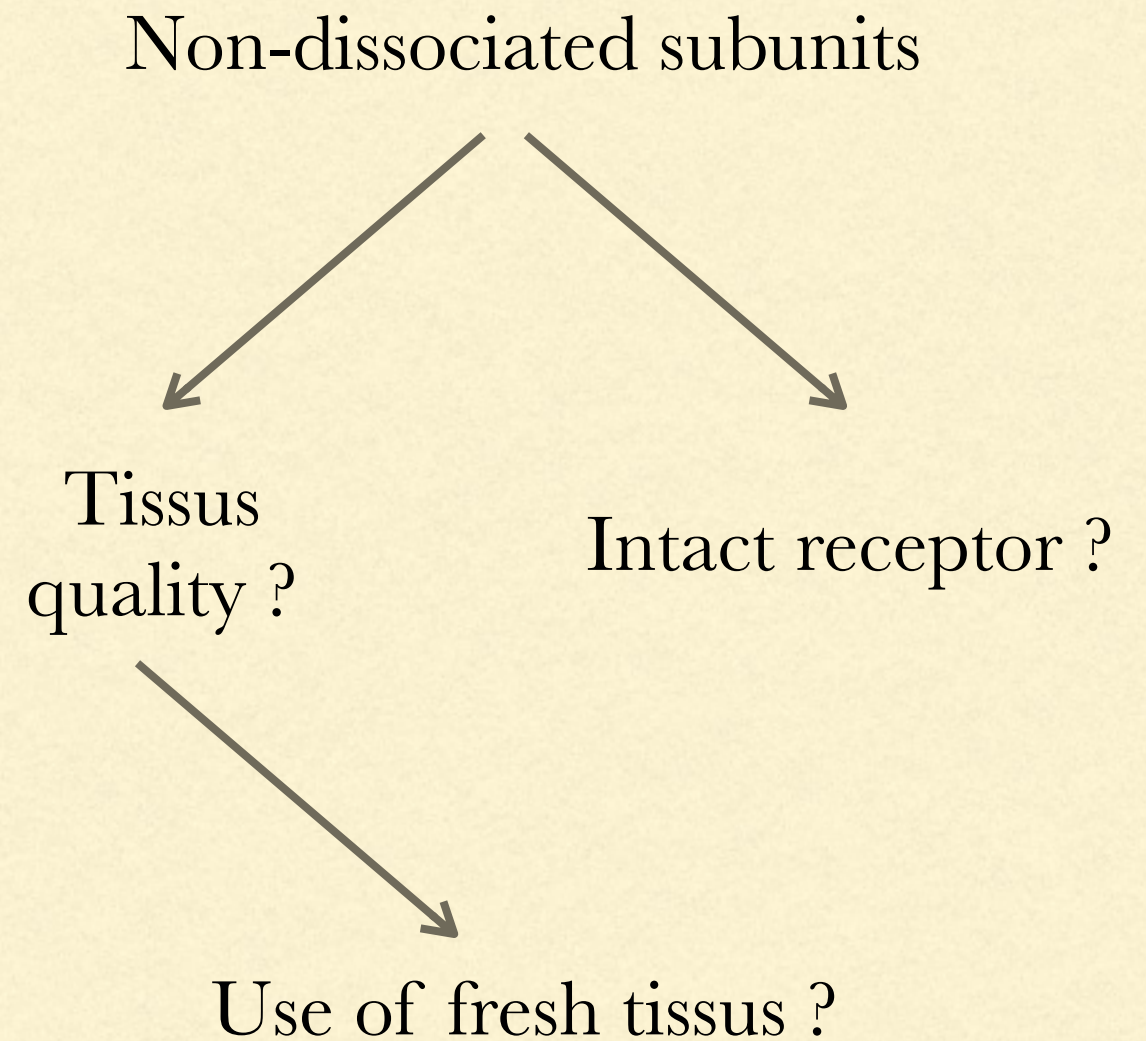
- Consensus sequence in green

- 46% of homology

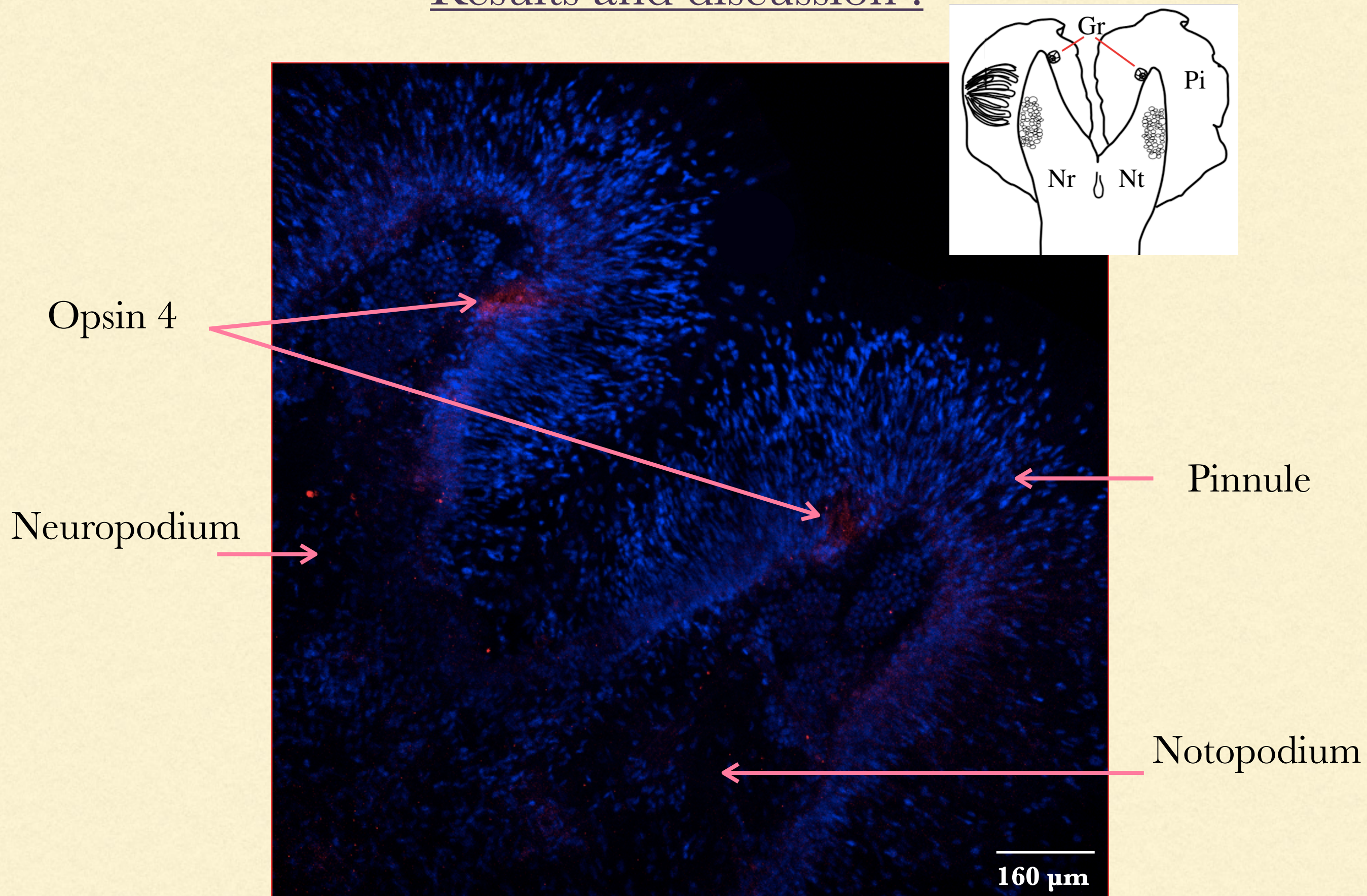
Results and discussion :



Western blot analysis of the $\alpha 7$ subunits of nicotinic receptors in *T. helgolandica* (extracts 1, 2, and 3).

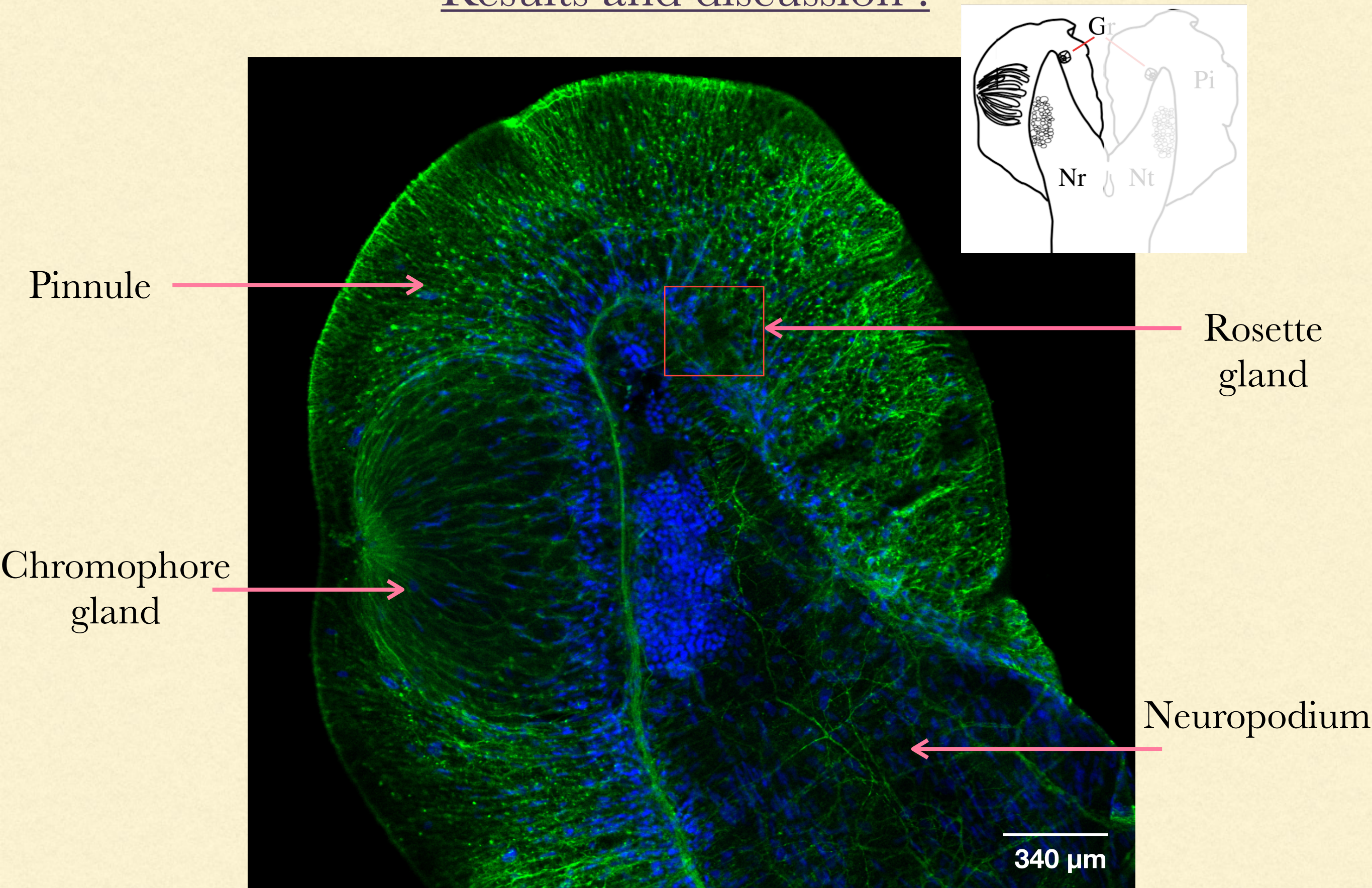


Results and discussion :



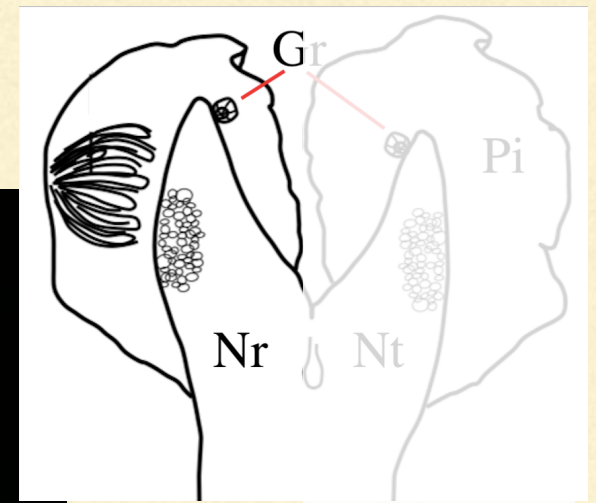
Immunofluorescent labeling of opsin 4 and nuclei

Results and discussion :



Immunofluorescent labeling of α -tubulin and nuclei

Results and discussion :

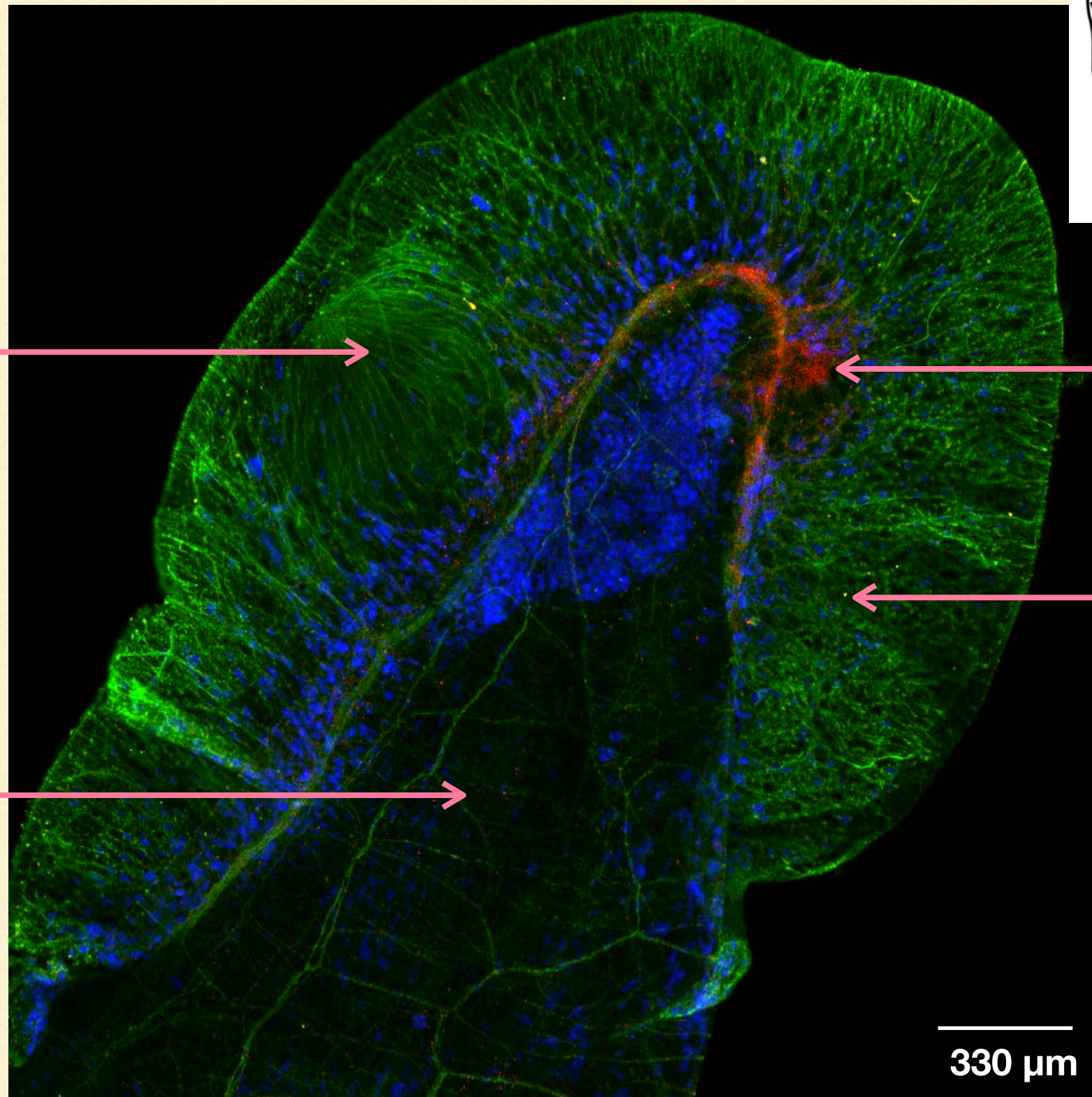


Chromophore
gland

Opsin 4

Pinnule

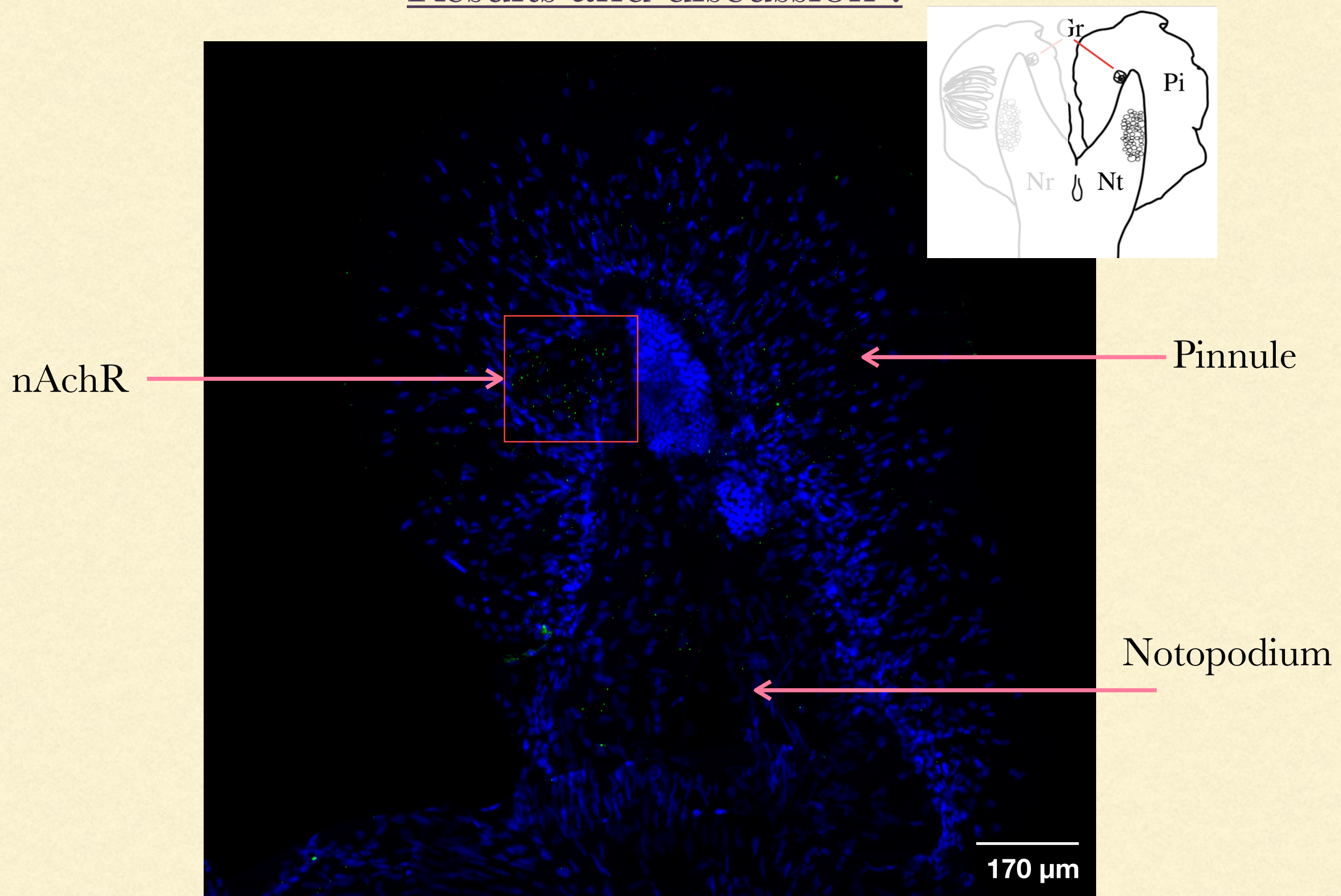
Neuropodium



330 μm

Double immunofluorescent labeling of α -tubulin,
opsin 4, and nuclei

Results and discussion :

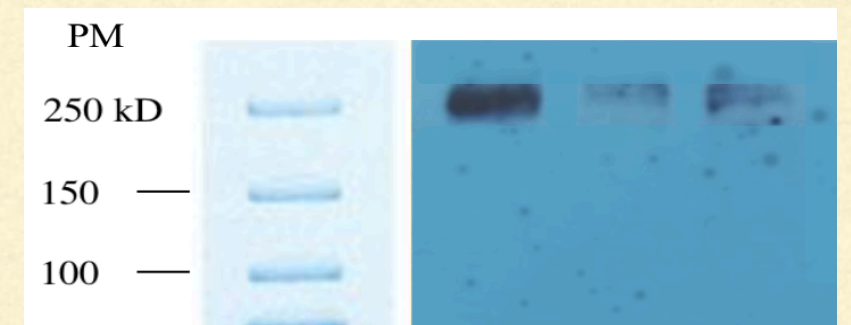


Immunofluorescent labeling of nAChRs and nuclei

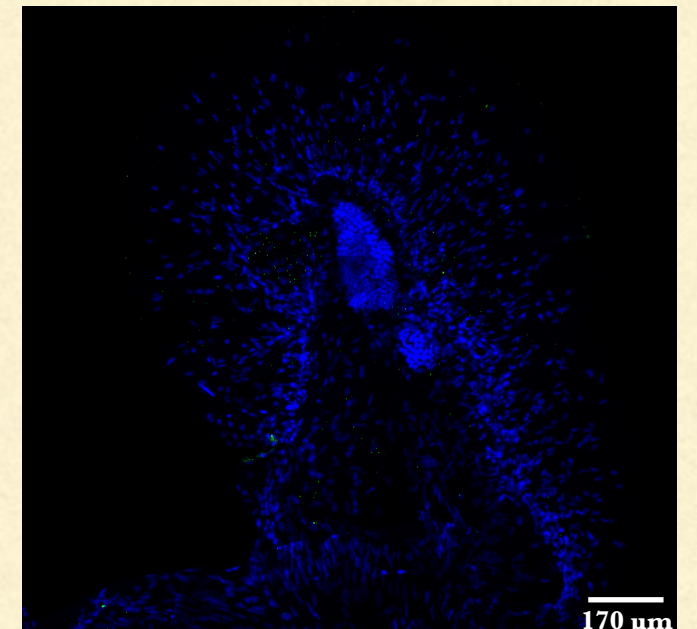
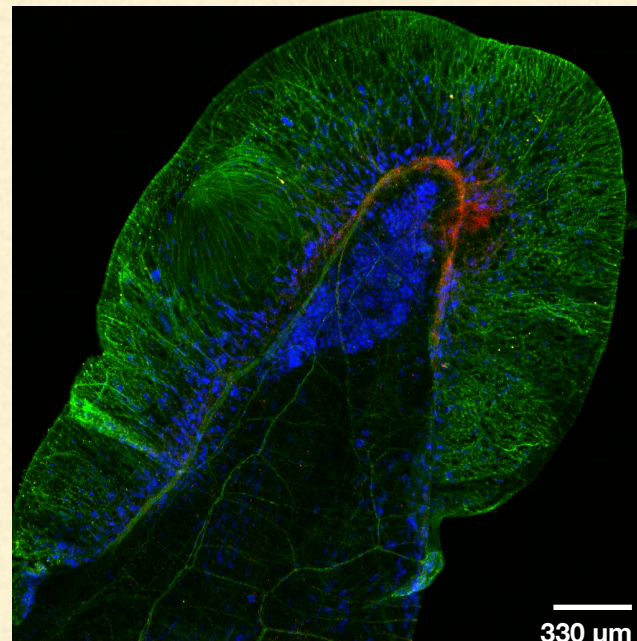
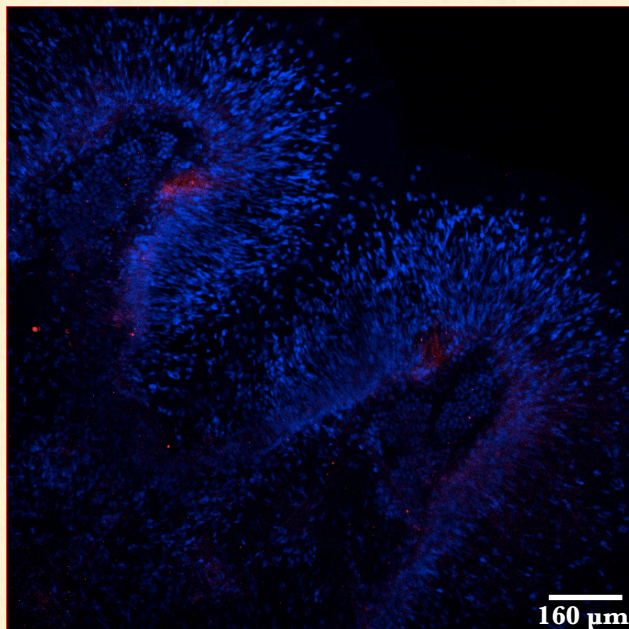
Conclusion :

- Presence of nearly all proteins involved in:
 - * Opsin 4 signaling cascade
 - * Nicotinic receptors signaling cascade

- Putative presence of the $\alpha 7$ subunits of nAChRs detected by Western blot

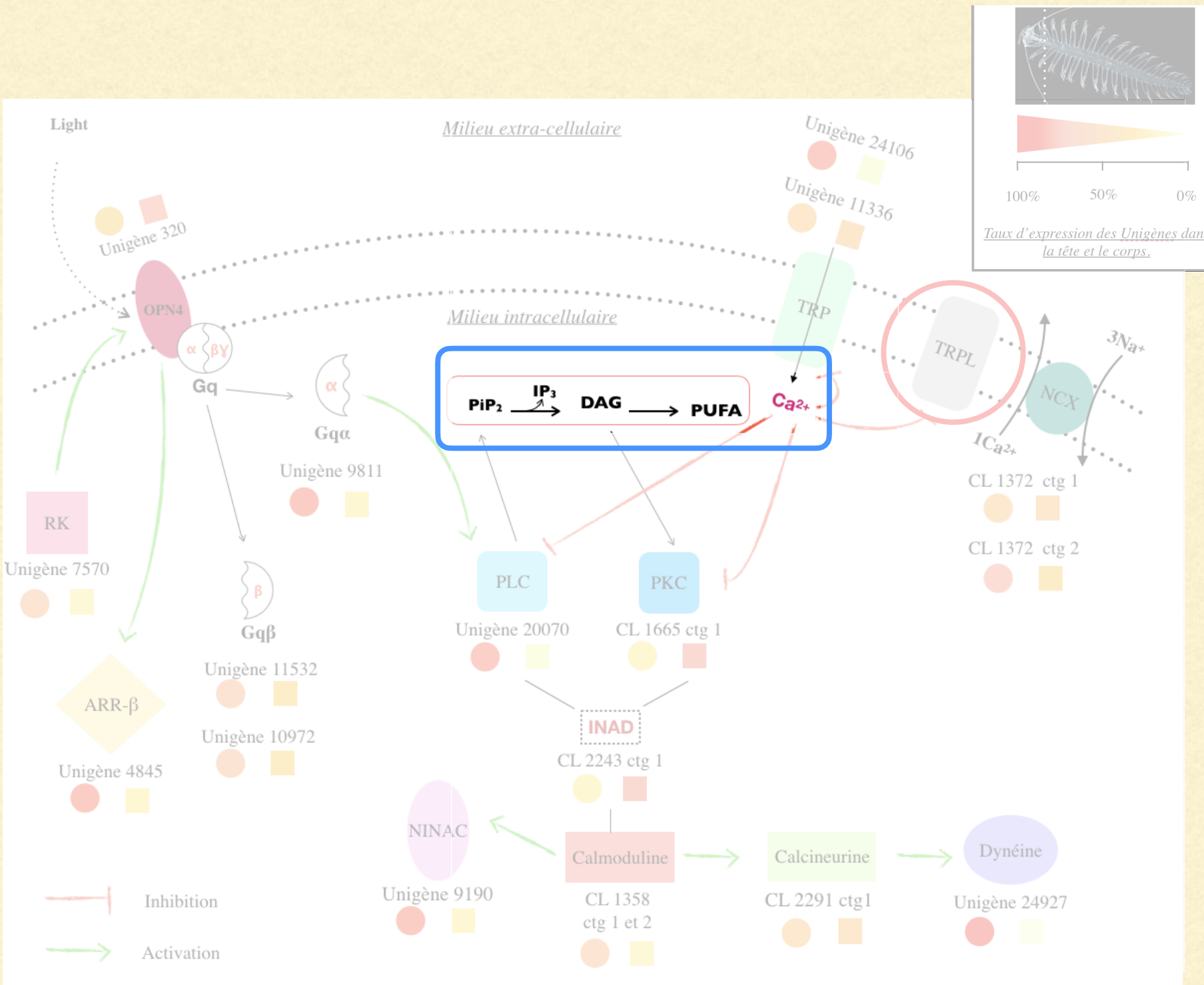


- Localization of opsin 4 and nAChRs in the light-emitting organ



Future Directions :

Opsin: Verify that opsin 4 absorbs in the yellow spectrum at 573 nm



Pharmacological tests:

induceur
inhibitor

Quantification :

- Second messengers :

IP3
DAG

- calcium

Proposed opsin 4 signaling cascade in *Tomopteris helgolandica*.

Future Directions :

Pharmacological tests:

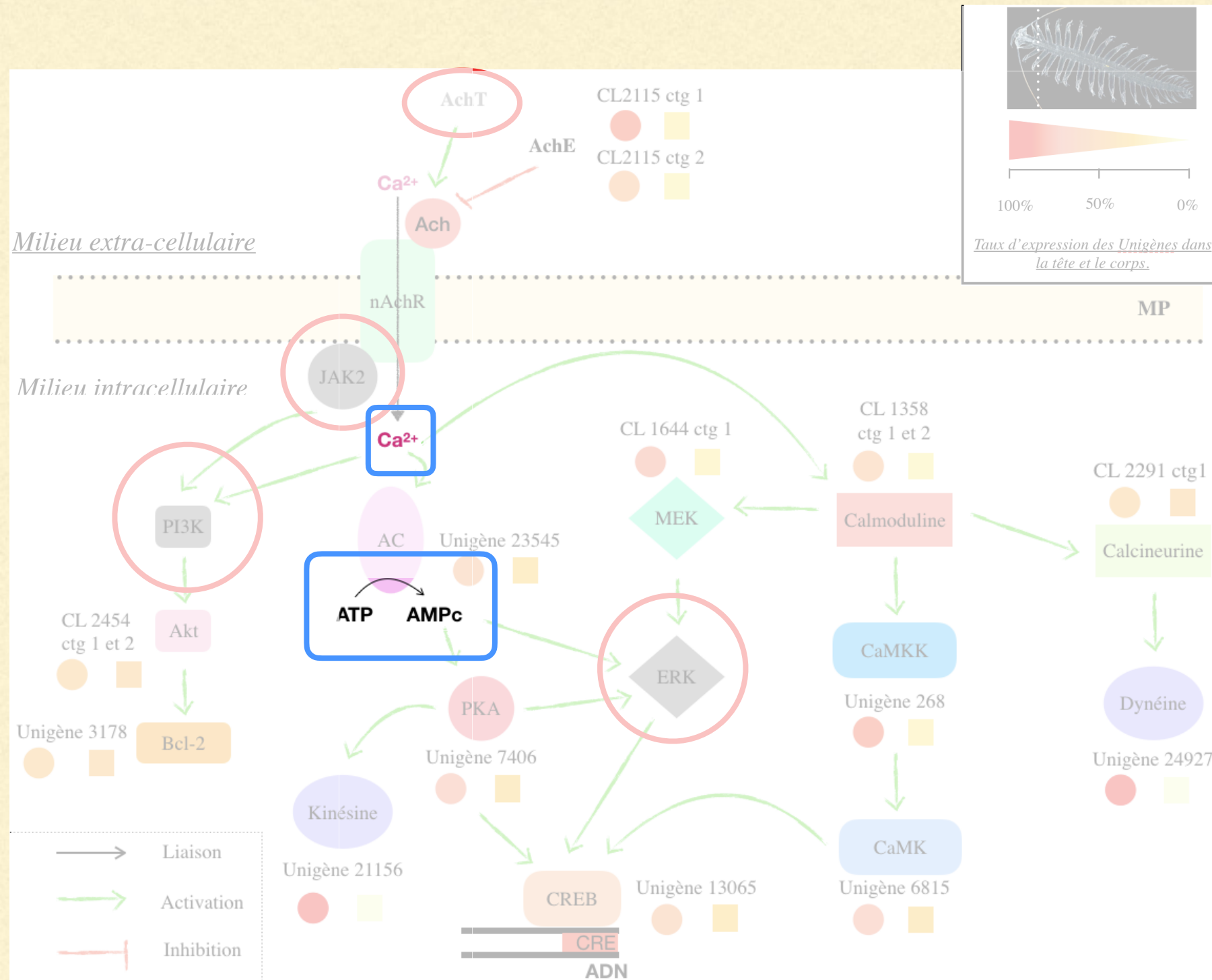
induceur
inhibitor

Quantification :

- Second messengers :

AMP_c

- calcium

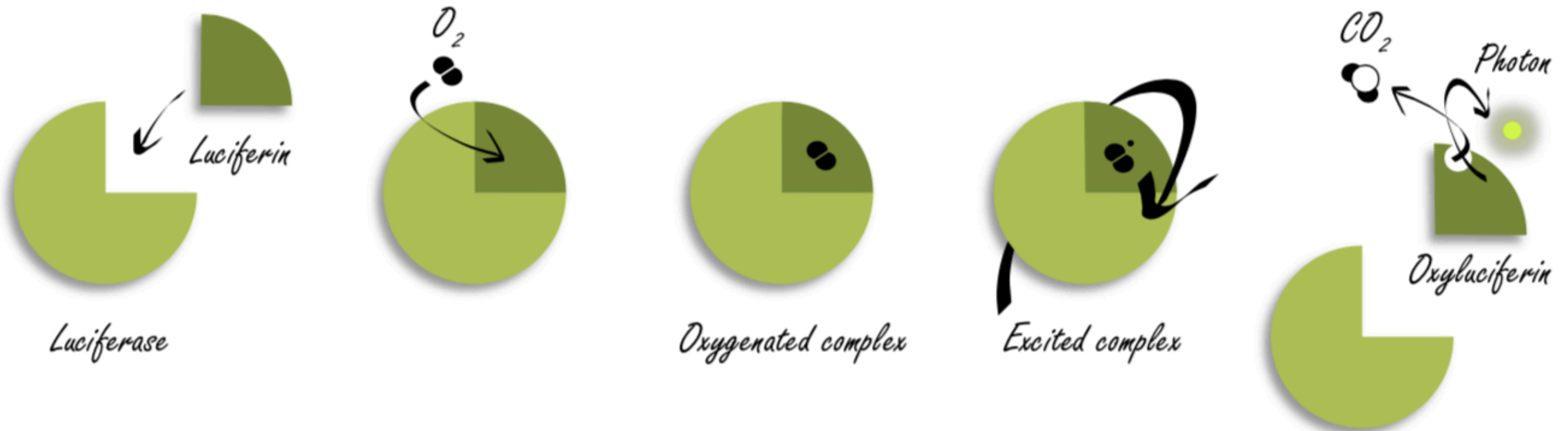


Proposed nicotinic receptor signaling cascade in
Tomopteris helgolandica.



Thank you for your
attention

Mécanismes de bioluminescence:



b

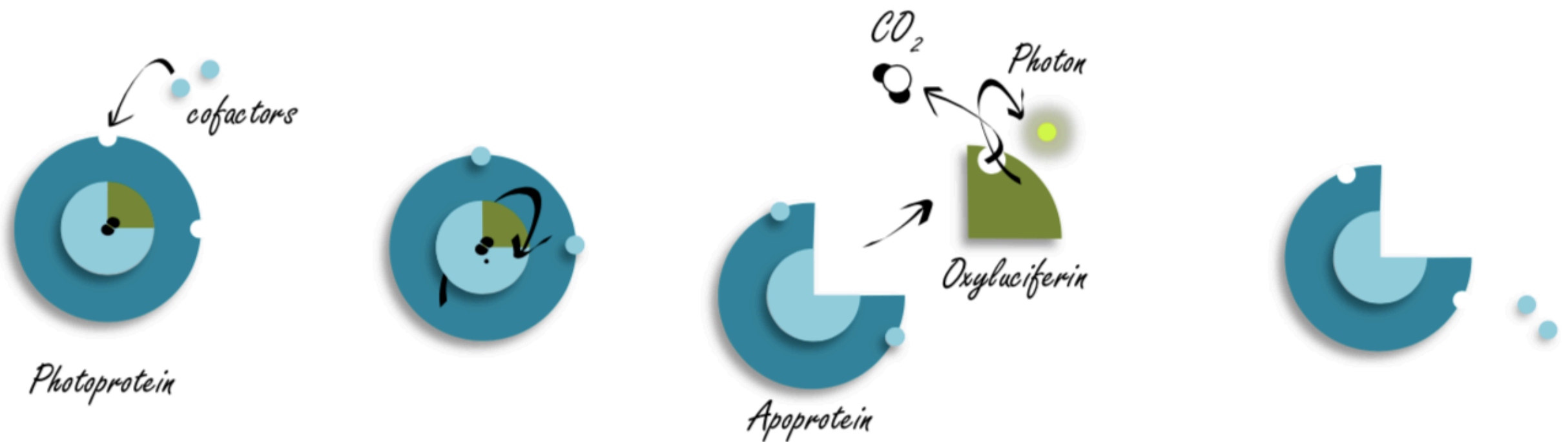


Schéma des deux mécanismes permettant la bioluminescence

(Gouveneaux, 2016)